

By 2nd Commutative Law of Propositions,
 $((x \in A \wedge \neg(x \in A)) \vee (\neg(x \in A) \wedge x \in B)) \vee ((\neg(x \in B) \wedge x \in A) \vee (x \in B \wedge \neg(x \in B)))$

By 2nd Negation Law of Propositions,
 $(F \vee (\neg(x \in A) \wedge x \in B)) \vee ((\neg(x \in B) \wedge x \in A) \vee F)$

By 1st Commutative Law of Propositions,
 $((\neg(x \in A) \wedge x \in B) \vee F) \vee ((\neg(x \in B) \wedge x \in A) \vee F)$

By 2nd Identity Law of Propositions,
 $(\neg(x \in A) \wedge x \in B) \vee (\neg(x \in B) \wedge x \in A).$

By Defn. of Not belongs to
 $(x \notin A \wedge x \in B) \vee (x \notin B \wedge x \in A)$

• We know that an element can either be a member of a set or not a member of a set. The following 4 cases arise:

1. $x \in A$ and $x \in B$. $\therefore \neg(x \in A)$ is False and $\neg(x \in B)$ is False, by the defn. of negation. By the defn. of ~~negation~~ not belongs to, $x \notin A$ and $x \notin B$ are both respectively False. By the defn. of AND, $(x \notin A \wedge x \in B)$ and $(x \notin B \wedge x \in A)$ are both False. By the defn. of OR, $(x \notin A \wedge x \in B) \vee (x \notin B \wedge x \in A)$ is False.
2. $x \notin A$ and $x \in B$. From Part 1, $x \notin B$ is False. By the defn. of not belongs to, $\neg(x \in A)$. By the defn. of negation, $\neg(\neg(x \in A))$ is False. By the Double Negation Law of Propositions, $x \in A$ is False. By the defn. of AND, $(x \notin A \wedge x \in B)$ is True. By the defn. of OR, $(x \notin A \wedge x \in B) \vee (x \notin B \wedge x \in A)$ is True.
3. $x \in A$ and $x \notin B$. From Part 1, $x \notin A$ is False. By the defn. of not belongs to, $\neg(x \in B)$. By the defn. of negation, $\neg(\neg(x \in B))$ is False. By the Double Negation Law of Propositions, $x \in B$ is False. By the defn. of AND, $(x \notin A \wedge x \in B) \vee (x \notin B \wedge x \in A)$ is True. By the defn. of OR, $(x \notin A \wedge x \in B) \vee (x \notin B \wedge x \in A)$ is True.
4. $x \notin A$ and $x \notin B$. From Cases 2 and 3, $x \in A$ and $x \in B$ are both False. By the defn. of AND, $(x \notin A \wedge x \in B)$ and $(x \notin B \wedge x \in A)$ are both False. By the defn. of OR, $(x \notin A \wedge x \in B) \vee (x \notin B \wedge x \in A)$ is False.